AS · Unit 2 · WBI12 Cells, Development, Biodiversity and Conservation

Plant structure and function

Plants are built differently from animals: their cells sit inside walls of cellulose, they store energy as starch, and they move water and food through specialised tissues. This page covers plant cell ultrastructure, starch and cellulose, the tissues of the stem and the fibres people make from them, the water and mineral ions plants need, and how drugs from plants are discovered and tested.

SPEC 4.1 Plant cell structure and how it compares with animal cells

WHAT THIS SPEC POINT ASKS

(i) Know the structure and ultrastructure of a plant cell (cell wall, chloroplast, amyloplast, vacuole, tonoplast, plasmodesmata, pits, middle lamella) and compare it with an animal cell. (ii) Know what each of these structures does.

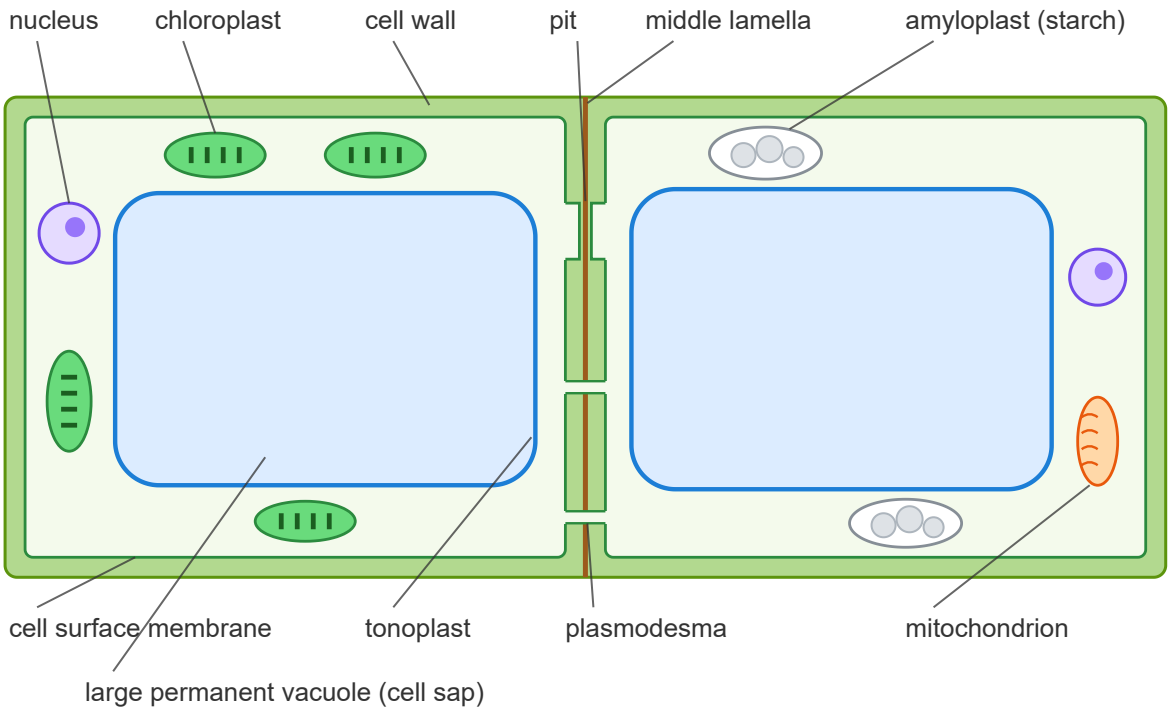
What plant and animal cells share

Both are eukaryotic cells, so both have a **nucleus** with a **nucleolus**, **rough and smooth ER**, a **Golgi apparatus**, **mitochondria**, **80S ribosomes** in the cytoplasm and a **cell surface membrane** (Topic 3). Plant cells add the structures below.

The extra structures in plant cells

Structure	What it is	What it does
Cell wall	Layer outside the cell surface membrane, made mainly of **cellulose** microfibrils in a matrix of other polysaccharides (pectins, hemicelluloses). Freely permeable.	Gives **support** and a fixed shape. Resists the outward pressure of the vacuole, so the cell becomes **turgid** instead of bursting.
Middle lamella	The thin layer between the walls of two neighbouring cells, made mainly of **calcium pectate**. It forms first, when a dividing cell lays down its new wall.	**Sticks neighbouring cells together**.
Plasmodesmata (singular plasmodesma)	Narrow channels through the walls of neighbouring cells, lined with cell surface membrane, so the	Let substances (water, ions, sucrose, signalling molecules) pass from cell to cell: **transport and**

Structure	What it is	What it does
	cytoplasm of one cell is continuous with the next.	**communication** between cells.
Pits	Thin areas of the wall where **no secondary thickening** (and no lignin) was laid down. They line up in pairs across neighbouring cells.	Let water and mineral ions move **sideways** (laterally), e.g. from one xylem vessel to the next or out to living cells.
Chloroplast	Lens-shaped organelle a few μm long with a **double membrane** (envelope). Inside: flattened sacs called **thylakoids**, stacked into **grana** and joined by membranes (lamellae), surrounded by a fluid **stroma** that holds enzymes, starch grains, circular DNA and 70S ribosomes. Chlorophyll is in the thylakoid membranes.	Site of **photosynthesis**: converts **light energy into chemical energy** (glucose).
Amyloplast	Colourless plastid with a double membrane, packed with **starch grains**; no thylakoids or chlorophyll. Common in storage organs (potato tubers, roots, seeds).	**Makes and stores starch**, and hydrolyses it back to glucose when the cell needs it.
Vacuole	Large, permanent sac filled with **cell sap**: water with dissolved sugars, mineral ions, pigments, waste and enzymes.	Water enters by osmosis, so the vacuole pushes the cytoplasm against the wall: **turgor**, which supports the plant. Also **stores** named substances (pigments, sugars, ions) and isolates waste or harmful chemicals.
Tonoplast	The **single, partially permeable membrane** around the vacuole, with carrier and channel proteins.	Controls what enters and leaves the vacuole.



Plant cell vs animal cell

Feature	Plant cell	Animal cell
Cell wall	**cellulose** wall outside the membrane	none
Middle lamella, plasmodesmata, pits	present	absent
Chloroplasts	in green tissues (not in root cells)	none
Carbohydrate store	**starch** grains, often in **amyloplasts**	**glycogen** granules
Vacuole	**large, permanent**, with a tonoplast	small, temporary vacuoles or vesicles only
Centrioles	absent (in flowering plants)	present
Lysosomes	rare (the vacuole does much of this job)	present
Cilia / flagella	absent (in flowering plants)	on some cells
Shape	fairly regular, fixed by the wall	variable

Feature	Plant cell	Animal cell
Both have	nucleus, nucleolus, rER, sER, Golgi apparatus, mitochondria, 80S ribosomes, cell surface membrane	

EXAM TIP

For "compare" questions, write paired sentences that name both cells: "a plant cell has a cellulose cell wall **whereas** an animal cell has none". A list of features with no cell named isn't credited.

MARK-SCHEME WORDING

- Animal cells only: "centrioles", "glycogen granules", "cilia", "flagella"; lysosomes accepted. Vesicles and temporary vacuoles are ignored.
- Chloroplast: "(site of) photosynthesis" / "converts light energy to chemical energy". "Makes food" is ignored.
- Starch / amyloplast: "store of {glucose / energy}". "Produces energy" and "stores carbohydrates" are ignored.
- Vacuole: "provides support (turgor)", "stores" + a **named** substance (water, sugars, mineral ions, pigments, enzymes). "Cell sap" alone and "gives the cell shape" are ignored.
- Plasmodesmata: "connect cells / allow communication between cells", "signalling substances pass through the cytoplasm (symplast)", or named molecules passing through.
- Pits: "allow (lateral) movement of water / mineral ions" between adjoining cells or out of the xylem.
- Walls with secondary thickening compared with walls without: "thicker / more layers", "contain lignin", "stronger / less flexible", "waterproof / impermeable".
- Drawings: amyloplasts and chloroplasts drawn **in the cytoplasm** and **smaller than the nucleus**; the tonoplast drawn as a **single** membrane, with the label line touching the membrane, not the middle of the vacuole.

Often asked as: MCQ on which structure stores starch (very common), name two structures found in animal but not plant cells, state the function of a named structure, draw and label amyloplast / chloroplast / tonoplast, compare a plant cell with a fungal cell.

COMMON MISTAKE

- Saying plant cells have no Golgi apparatus: they do (WBI12 Oct 2024 report).
- Putting chloroplasts in a **root** cell. Roots are underground, so their cells have none.
- Writing "cell sap" or "nutrients" as the vacuole's contents, or "nutrients" for what passes through plasmodesmata. Name the substances.
- Mixing up **pits** (thin areas of the wall) with **plasmodesmata** (cytoplasmic channels through the wall).
- Calling the vacuole the tonoplast. The tonoplast is the membrane around it.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Fungal cells** compared with plant cells: fungal walls are made of **chitin**, not cellulose; fungi store **glycogen**, not starch; they have no chloroplasts and no plasmodesmata (WBI12 Jan 2021).
- How a plant cell **gets bigger** after division: it takes up water into the vacuole and makes more cytoplasm, organelles and proteins, plus new membrane and wall. Describing division instead loses the marks (WBI12 Oct 2021).
- Sphere and hemisphere volume in plant contexts ($4/3\pi r^3$, halved for a bowl shape) (WBI12 Jun 2022).

SPEC 4.2 Recognising plant organelles in electron micrographs

WHAT THIS SPEC POINT ASKS

Identify the plant cell structures from 4.1 in electron microscope images.

What each structure looks like in a TEM

Structure	How to spot it
Cell wall	Thick, fairly even grey band around the outside of the cell. Between two cells you may see a thin darker line down the middle: the **middle lamella**. The cell surface membrane sits tight against the inside of the wall, so it is hard to see.
Plasmodesmata	Thin dark channels running straight across the wall, joining the cytoplasm of two cells.
Pits	Gaps or thin places in a thick (lignified) wall, often in pairs on either side of the wall.
Chloroplast	Large oval with a double membrane. Inside: **stacks of thin parallel lines** (grana) joined by single lines, in a grainy stroma. Often has pale starch grains and small dark lipid droplets. Bigger than a mitochondrion.
Amyloplast	Double membrane, filled with one or more large **starch grains** (sometimes showing rings). **No thylakoids.**
Vacuole	Large pale area that looks empty, bounded by one thin membrane (the **tonoplast**). Often pushes the cytoplasm into a thin layer at the edge.
Mitochondrion (to compare)	Smaller, double membrane, inner membrane folded into **cristae**, not stacked discs.

EXAM TIP

Chloroplast or mitochondrion? Look inside. **Stacked thylakoids** = chloroplast. **Folds of the inner membrane** (cristae) = mitochondrion. Chloroplast or amyloplast? Starch grains with **no** thylakoid stacks = amyloplast.

A large pale vacuole, a cell wall or chloroplasts tells you the image is a plant cell.

WORKED EXAMPLE

Worked example: actual size from a micrograph

A chloroplast measures 42 mm long on a micrograph at × 6000.

Convert to µm: 42 mm × 1000 = 42 000 µm.

Actual size = image size ÷ magnification = 42 000 ÷ 6000 = **7.0 µm**.

With a scale bar instead: measure the bar in mm, convert to µm, and divide by the value written on the bar to get the magnification first.

MARK-SCHEME WORDING

- Amyloplast: spelling is marked leniently, but "starch grain" is **wrong** when the organelle is asked for.
- Plasmodesma: if you name pits, the cell wall or the middle lamella instead, you can still get the function mark (ECF) when the function matches.
- Vacuole: a label line to the tonoplast is accepted.

Often asked as: name the structure labelled X (amyloplast in cassava, plasmodesmata between two cells, a chloroplast, pits in a xylem wall, a mesophyll vacuole), then give its function; magnification or actual-size calculations from the image.

COMMON MISTAKE

- Not recognising plasmodesmata: a significant minority missed them (WBI12 Oct 2020 report).
- In magnification sums, measuring the specimen when the question wants the scale bar, or forgetting to convert mm to µm (× 1000).

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Examined in past papers even though the spec doesn't list it.

Recognising a **pit** as a gap in a lignified xylem wall in a micrograph, and saying it allows sideways movement of water (WBI12 Oct 2023).

SPEC 4.3 Starch and cellulose

WHAT THIS SPEC POINT ASKS

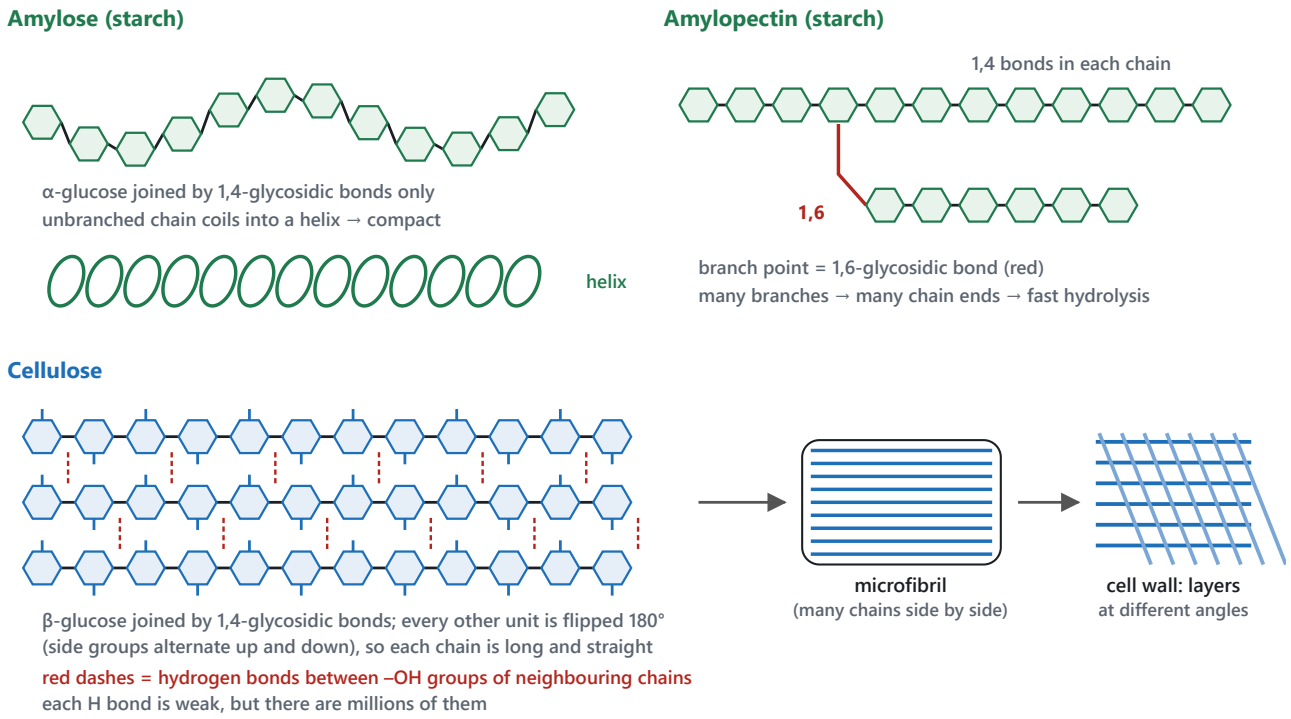
Know the structure and function of starch and cellulose, including how hydrogen bonds between β-glucose molecules form cellulose microfibrils.

Starch: the plant's energy store

- Starch is a mixture of two polymers of **α-glucose** (Unit 1, 1.2):
  - **Amylose**: **1,4-glycosidic bonds** only, so an unbranched chain that coils into a **helix**. Very compact.
  - **Amylopectin**: 1,4 bonds **and 1,6-glycosidic bonds** at the branch points, so **branched**.
- It is stored as grains in **amyloplasts** and in chloroplasts.
- Why it is a good store, as feature → consequence pairs:
  - Made of **glucose** → hydrolysed to glucose for **respiration**.
  - **Compact** (coiled, branched) → a lot of glucose stored in a small space.
  - **Insoluble** → **doesn't affect the water potential** of the cell, so no osmotic water uptake; and it can't diffuse out.
  - **Large** → can't cross the membrane, so it stays in the cell.
  - **Branched** (amylopectin) → many chain ends where enzymes can work at once → **rapid hydrolysis** to glucose.

Cellulose: the structural polysaccharide

- Cellulose is a polymer of **β-glucose**. In β-glucose the –OH on carbon 1 points **up** (above the ring); in α-glucose it points down.
- To join carbon 1 of one β-glucose to carbon 4 of the next by a **1,4-glycosidic bond**, **every other molecule is turned over (rotated 180°)**.
- That flipping makes each chain **long, straight and unbranched**, with –OH groups sticking out on both sides.
- The –OH groups of neighbouring parallel chains form **hydrogen bonds** with each other (δ+ H of one –OH attracted to the δ− O of another). One hydrogen bond is weak, but there are huge numbers of them, so together they hold the chains firmly side by side.
- Bundles of many chains held together like this form a **microfibril**. Microfibrils are very strong in tension (high **tensile strength**), insoluble, and hard to break down.
- In the cell wall, microfibrils lie in a matrix of other polysaccharides and are laid down in layers (4.4).



Starch (α-glucose) coils or branches; cellulose (β-glucose) forms straight chains held side by side by hydrogen bonds, which bundle into microfibrils.

Starch vs cellulose vs glycogen

	Starch (amylose + amylopectin)	Cellulose	Glycogen
Monomer	α-glucose	**β-glucose**	α-glucose
Glycosidic bonds	1,4 (amylose); 1,4 and 1,6 (amylopectin)	**1,4 only**	1,4 and 1,6
Branching	amylose none; amylopectin branched	**none**	**highly** branched
Shape	helix (amylose); branched (amylopectin)	**straight** chains; alternate units flipped 180°	compact, branched granule
Hydrogen bonds	within the amylose helix	**between neighbouring chains** → microfibrils	within the molecule
Soluble?	no	no	no
Job	**energy store** in plants	**structural**: cell walls	**energy store** in animals and fungi

	Starch (amylose + amylopectin)	Cellulose	Glycogen
Where	amyloplasts, chloroplasts	plant cell walls	liver and muscle cells
Broken down by	amylase (1,4 bonds) and other enzymes	**cellulase** (most animals don't make it)	enzymes in liver and muscle cells

EXAM TIP

"Compare starch and cellulose" needs at least one **similarity** as well as differences, written as pairs:

- Both are **polysaccharides / polymers of glucose** joined by **(1,4) glycosidic bonds**.
- Starch is made of **α**-glucose **whereas** cellulose is made of **β**-glucose.
- Starch has 1,4 **and** 1,6 bonds (branched amylopectin) **whereas** cellulose has 1,4 bonds only (unbranched).

MARK-SCHEME WORDING

- Starch for storage: "polymer of / contains glucose" (glucose is the energy source); "branched / 1,6 bonds, **therefore** can be rapidly hydrolysed"; "compact, so more {glucose / energy} stored"; "insoluble, therefore does not affect osmosis". "Easy to break down" is ignored.
- Starch: α-glucose; amylose and amylopectin; 1,4 **and** 1,6 bonds. β is rejected.
- Cellulose: "β-glucose (alternate molecules inverted)", "1,4 glycosidic bonds only", "hydrogen bonds between (adjacent) cellulose molecules", "forms microfibrils".
- How cellulose makes a wall strong: "hydrogen bonds between adjacent chains", "{layers / sheets} of microfibrils", "microfibrils at different angles" (criss-cross or mesh accepted).
- Compare: full marks need a similarity. Separate paragraphs on each molecule are **not pieced together**.

Often asked as: MCQ on starch or cellulose statements; identify a cellulose chain or a branched starch from drawings; compare cellulose and starch (3 marks); explain how starch suits energy storage; explain how cellulose and microfibrils strengthen the wall, using a micrograph.

COMMON MISTAKE

- Confusing a **cellulose molecule** (one chain) with a **microfibril** (many chains hydrogen-bonded together) (WBI12 Oct 2020 report).
- Writing "easier to break down" instead of "**faster hydrolysis**" (WBI12 Oct 2021 report).
- Giving "insoluble" when the question asks why a store holds a lot of **energy**. That needs "polymer of glucose" and "compact" (WBI12 Jun 2024).
- Saying animals can't digest cellulose because it has 1,4 bonds. Starch has 1,4 bonds too. The difference is **β**-1,4 bonds, which need **cellulase**.

- Writing about pectin, hemicellulose or lignin when the question asks only about cellulose and microfibrils.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Recognising **β-glucose** in a structural formula (–OH on carbon 1 above the ring; alternate units inverted in a chain) (WBI12 Jan 2020).
- Cellulose is in the walls of xylem, phloem **and** sclerenchyma (WBI12 Oct 2020).
- Picking out the **branched** (amylopectin) drawing and naming the 1,6 bond (WBI12 Jun 2023).

SPEC 4.4 Microfibrils, secondary thickening and the strength of plant fibres

WHAT THIS SPEC POINT ASKS

Explain how the way cellulose microfibrils are arranged, plus secondary thickening, gives xylem vessels and sclerenchyma fibres the physical properties that make plant fibres useful to humans.

How a cell wall is built

- **Primary wall**: laid down first, while the cell is still growing. Thin and flexible; microfibrils form a loose **mesh** in a matrix of pectins and hemicelluloses, so the wall can stretch.
- **Secondary wall**: added **inside** the primary wall once the cell stops growing (only in some cells, e.g. xylem vessels and sclerenchyma fibres). Microfibrils are laid down in several **layers**, and the microfibrils in each layer run at a **different angle** from the layer next to it.
- **Secondary thickening** = this extra secondary wall, usually impregnated with **lignin**.
- **Lignin** is a complex polymer that fills the spaces between microfibrils. It makes the wall **rigid** (resists compression and stops the cell collapsing) and **waterproof** (impermeable). The cell contents die, leaving a hollow tube.
- Where lignin is missing, **pits** remain, so water can still move sideways.
- In xylem, lignin is often laid down as **rings, spirals** or a net. This lets young xylem stretch as the stem grows and the stem bend without the tube collapsing.

Why this gives strength

Feature	Physical property it gives
Many **hydrogen bonds** between cellulose chains	each microfibril has high **tensile strength** (hard to pull apart)

Feature	Physical property it gives
Layers of microfibrils at different angles (criss-cross)	strong **in many directions**, not just along one line
Lignin between microfibrils	**rigid**, resists compression, **waterproof**; walls don't collapse under the tension in the xylem
Long cells overlapping end to end	long, strong **fibres** that can be extracted
Gaps between microfibrils (and pits)	water and ions can still pass through the wall

How humans use plant fibres

Fibre	What it is	Uses
Cotton	seed hairs: almost pure cellulose, no lignin	cloth (textiles)
Flax (linen), **hemp**, **jute**	**sclerenchyma fibres** from the stem	cloth, rope, sacking, fibre composites
Sisal	fibres from the leaves	rope, twine, matting
Wood	mostly xylem and fibres	timber, paper (pulped wood)

Fibres are separated from the rest of the stem by **retting**: microorganisms (or chemicals) break down the pectin of the middle lamella, while the cellulose and lignin of the fibres resist.

MARK-SCHEME WORDING

- Xylem wall properties: "cellulose molecules organised into microfibrils"; "layers of microfibrils at different angles / a mesh"; "{hydrogen bonds / lignin / secondary thickening} give {strength / support}"; "microfibrils held together by {pectin / pectate}"; gaps let water and ions through.
- Lignin: "makes the wall impermeable to water (waterproof)", "provides strength", laid down as rings or spirals.
- Xylem for **support**: "walls have secondary thickening / lignin" + strength. For **transport**: "hollow tubes with no {end walls / cytoplasm}" + "no obstruction to the water column".
- Climbing plant v sunflower: the climber "is supported by the pole" and "needs to be **flexible** to bend round it", so needs less lignin; the sunflower must hold itself up.

Often asked as: explain the physical properties of a xylem vessel wall; role of secondary thickening in sclerenchyma; why a fibre such as jute is strong; interpret cellulose and lignin data (bamboo with age).

COMMON MISTAKE

- Leaving out "lignified walls / secondary thickening" in support answers (WBI12 Jan 2020 report).
- Writing "hollow" without saying "no end walls / no cytoplasm".
- Giving **functions** when **physical properties** are asked (WBI12 Oct 2023), or treating lignin as something that carries water rather than giving strength and waterproofing (WBI12 Jan 2024).
- Saying cellulose fibres resist **compression**. Cellulose gives **tensile** strength; lignin resists compression.
- Saying lignin strengthens the cell **membrane**. It is in the **wall** (WBI12 Oct 2019).

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- **Bamboo**: as the stem ages, the % cellulose falls and the % lignin rises; ratio cellulose : lignin about 2 : 1 at one year (WBI12 Jun 2025).
- **Jute** has high tensile strength because of its high cellulose and lignin content, and because its fibres are twisted or woven (WBI12 Oct 2025).
- **Tracheids** (long xylem cells with tapering ends) are accepted as support cells like sclerenchyma (WBI12 Jun 2021).
- **Capillary action** in the narrow xylem lumen helps water rise (WBI12 Jan 2020).

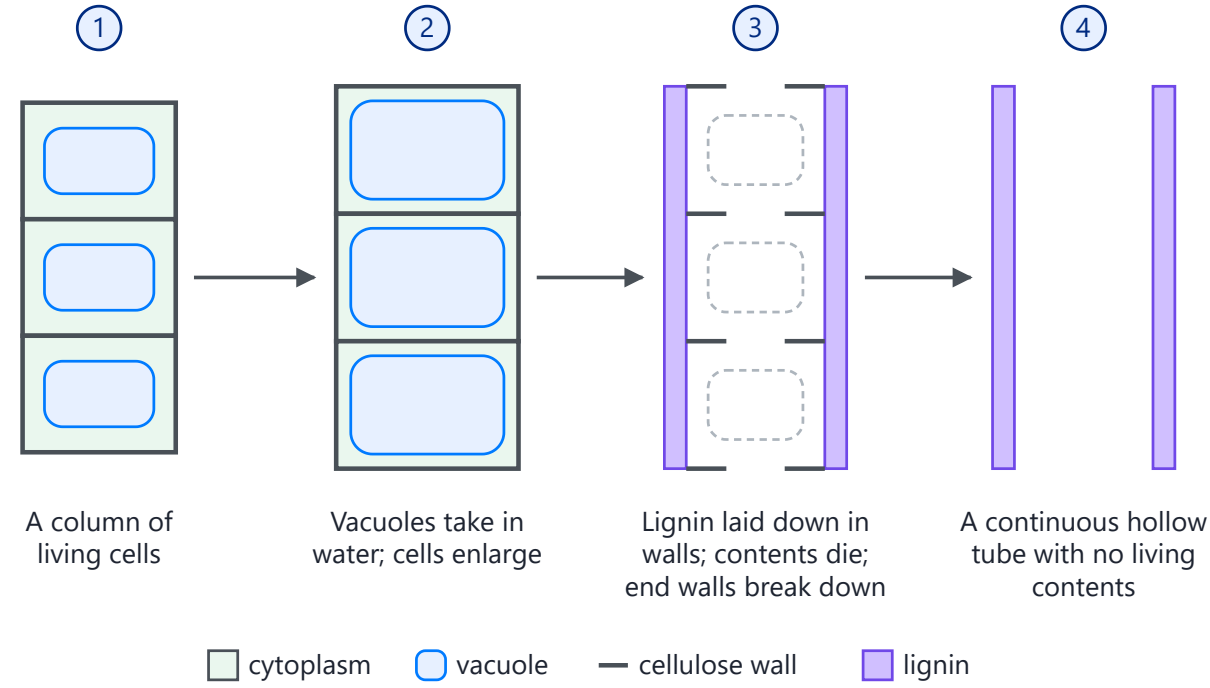
SPEC 4.5 Sclerenchyma, xylem and phloem

WHAT THIS SPEC POINT ASKS

Know how sclerenchyma fibres (support), xylem vessels (support, and transport of water and mineral ions) and phloem (translocation of organic solutes) are alike and different in structure, where they sit in the stem, and what they do.

Xylem vessels

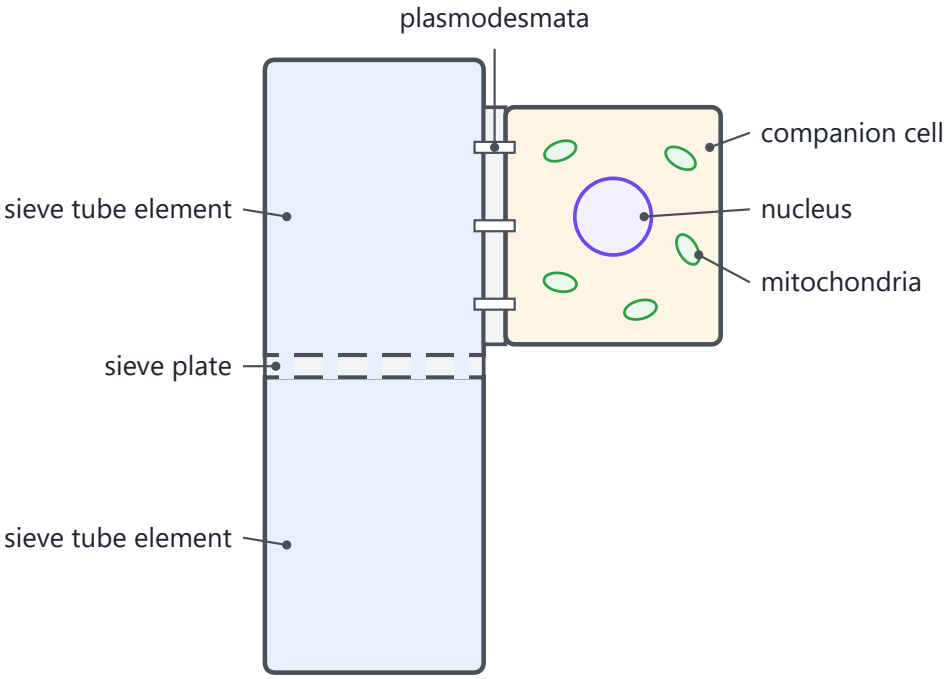
- Formed from a column of cells joined end to end. Their **end walls break down** and their contents die, leaving a **continuous hollow tube** with **no cytoplasm**.
- Walls are **lignified** (secondary thickening in rings, spirals or a net) → support, waterproof, don't collapse under tension.
- **Pits** in the walls let water move sideways between vessels.
- Carry **water and mineral ions** from the roots **up** to the leaves (one direction) in the transpiration stream; also **support** the plant.



How a xylem vessel forms from a column of living cells.

Phloem

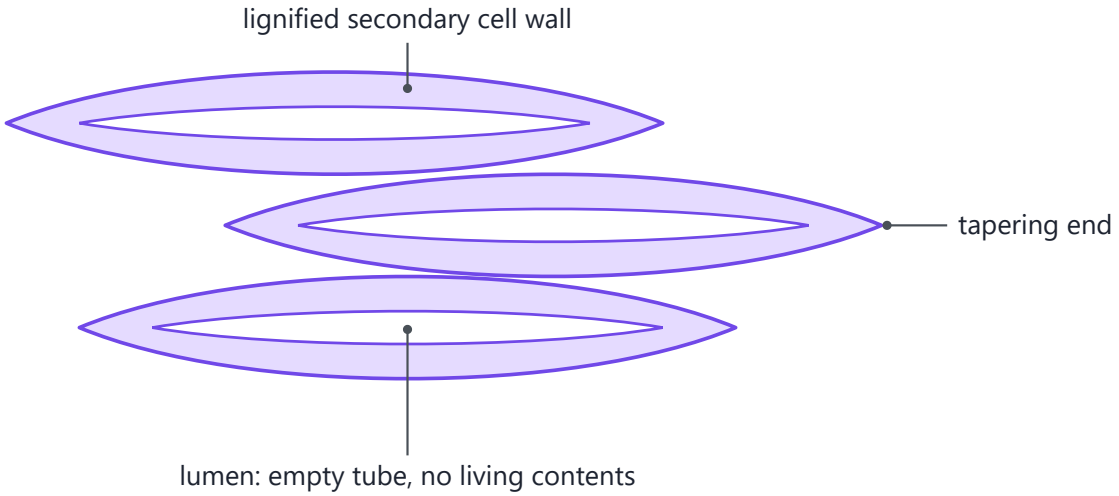
- **Sieve tube elements**: **living** cells joined end to end. The end walls are perforated to form **sieve plates**. Each has **no nucleus**, very few organelles and only a thin layer of cytoplasm, so sap can flow through.
- Walls are **cellulose only, not lignified**.
- **Companion cells** lie beside each sieve tube element, linked to it by **plasmodesmata**. They have a nucleus, dense cytoplasm and **many mitochondria**: they supply the ATP for **active loading of sucrose** and keep the sieve tube element alive.
- Phloem carries **organic solutes**, mainly **sucrose** and **amino acids**, from **sources** (e.g. leaves) to **sinks** (growing points, roots, storage organs, fruits). This is **translocation**, and it goes **both up and down** the plant.



Sieve tube elements and a companion cell.

Sclerenchyma fibres

- Long, narrow cells with **tapering, closed ends** (end walls), overlapping in bundles.
- Very thick **lignified secondary walls**, a small empty lumen, some simple pits. **Dead** when mature.
- **Support only**: they don't transport anything.



Sclerenchyma fibres: dead, hollow cells with thick lignified walls and tapering ends.

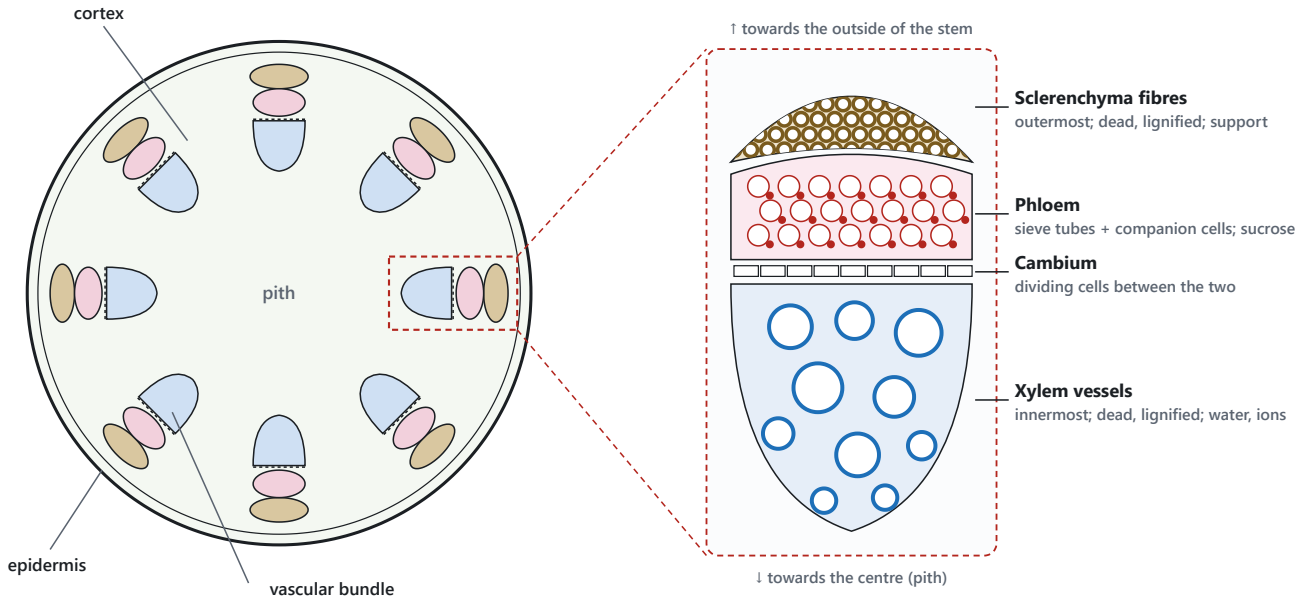
Comparison table

	Sclerenchyma fibre	Xylem vessel	Phloem sieve tube
Living or dead	dead	dead	**living** (with companion cells)
Contents	none (hollow)	none (hollow)	thin cytoplasm, **no nucleus**
End walls	**present** (tapered, closed ends)	**absent** (continuous tube)	**sieve plates** (perforated end walls)
Lignin / secondary thickening	yes, very thick	yes (rings, spirals, net)	**no**
Wall material	cellulose + lignin	cellulose + lignin	cellulose
Links to other cells	pits	**pits**	**plasmodesmata** (to companion cells)
Function	**support**	**water and mineral ions** (upwards only) + support	**sucrose, amino acids** (both directions)
Position in a stem	**outermost** of the three, a cap on the outside of each vascular bundle	**inner** side of the vascular bundle, towards the centre	**outer** side of the vascular bundle, between xylem and sclerenchyma

Similarities to remember: all three are long, narrow cells with **cellulose** walls. Xylem vessels and sclerenchyma fibres are both **dead, hollow, lignified** and give **support**, and both have **pits**. Xylem and phloem are both **tubes for transport** in the vascular bundles, and neither a mature xylem vessel nor a sieve tube element has a **nucleus**.

Where they are

- **Stem**: vascular bundles in a **ring** near the outside. In each bundle, from outside to inside: **sclerenchyma** → **phloem** → **cambium** → **xylem**. Pith in the centre.
- **Root**: xylem in the **centre**, often star-shaped, with phloem in the gaps between its arms (all inside the endodermis). A central core resists pulling forces as the root anchors the plant.
- **Leaf**: in each vein and the midrib, **xylem towards the upper surface** and **phloem towards the lower surface**.



Stem in transverse section. In each vascular bundle the sclerenchyma is outermost, then phloem, cambium and, nearest the centre, xylem.

MARK-SCHEME WORDING

- Phloem: "{transport / translocation} of {sucrose / amino acids}" "from {source / leaves} to {sink / site of use or storage}". "Glucose", "nutrients" and "sugar" are **ignored**.
- Xylem: "transport of water and mineral ions" and support. Sclerenchyma: "support" ("strength" and "rigidity" alone are ignored).
- Xylem v sclerenchyma: similarities "both contain {cellulose / pits / lignin / dead cells}", "both hollow / no cytoplasm". Difference: "sclerenchyma has end walls (closed, pointed ends) **whereas** xylem does not".
- Phloem v xylem (structure): similarities "both have cellulose walls", "both tubular", "both have no nucleus". Differences: "phloem has sieve plates", "phloem contains cytoplasm, so not hollow", "phloem has no lignin / secondary thickening", "phloem has plasmodesmata whereas xylem has pits"; living v dead accepted.
- Phloem adapted: "continuous tube / sieve plates let the solution flow", "plasmodesmata", "companion cells provide ATP for active loading of sucrose".
- Root v stem: "in the stem xylem and phloem are in separate vascular bundles arranged in a ring; in the root they are in a central group". Use comparative language and say which tissue.

Often asked as: label a stem section (sclerenchyma, phloem, cambium, xylem); compare xylem with sclerenchyma or with phloem; describe tissue positions in root, stem and leaf; how phloem structure suits its job; dye taken up by a celery stalk; a parasitic plant or a millipede feeding from the vascular tissue.

COMMON MISTAKE

- Swapping **sclerenchyma** and **phloem** when labelling, or letting a label line touch the wrong tissue (WBI12 Jan 2023, Jan 2020 reports).

- Thinking both xylem and phloem are lignified, or that xylem has end walls or sieve plates (WBI12 Oct 2019, Jun 2022 reports).
- Writing "glucose" or "nutrients" for what phloem carries. It is **sucrose** (and amino acids).
- Giving functions when the question asks for **structure** (WBI12 Jun 2022 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Phloem moves sap **both up and down**; xylem moves water **upwards** only (WBI12 Jan 2021).
- **Calcium ions** travel up the plant in the **xylem** (WBI12 Jan 2021).
- A parasitic plant grows into the host's **phloem** and absorbs **sucrose** by diffusion, facilitated diffusion or active transport (**not** osmosis) (WBI12 Jun 2023).
- A dye placed in water travels up a celery stalk in the **xylem**, so only the xylem is stained (WBI12 Jun 2023).
- Recognising **parenchyma** (thin-walled living packing cells) in a stem section (WBI12 Jan 2021).

SPEC 4.6 Core Practical 7: plan diagrams and cell drawings of plant tissues

WHAT THIS SPEC POINT ASKS

Use a light microscope to observe transverse sections of roots, stems and leaves; draw and label plan diagrams of them; draw and label cells of plant tissues; and identify sclerenchyma fibres, phloem, sieve tubes and xylem vessels and where they are.

Plan diagram or cell drawing?

	Plan diagram (low power)	Cell drawing (high power)
Shows	**outlines of tissues** only: where each tissue starts and ends	a **few individual cells** of one tissue, with their walls (and nucleus etc. if visible)
Never shows	individual cells	the whole section
Use it to	show the **position** of xylem, phloem, sclerenchyma	show the **structure** of e.g. xylem vessels or sieve tubes

CORE PRACTICAL

Core practical 7: method in brief

1. Wet a razor blade or scalpel and cut **very thin** transverse sections of a stem (or root) on a white tile, cutting away from your fingers. For a leaf, hold it between two pieces of polystyrene or carrot and shave thin slices.
2. Move the sections with a soft paintbrush into water in a watch glass. Choose the **thinnest** one: a thin partial section is better than a thick whole one (thick sections let little light through and cells overlap).
3. Put it on a slide and add a drop or two of stain, e.g. **toluidine blue O** for 1–2 minutes, then rinse or blot off the excess.
4. Lower a **coverslip** at an angle from one side to avoid air bubbles.
5. Focus on **low power** first and draw a **plan diagram** of the tissue layout. Then switch to **high power** and draw a few cells of one tissue.
6. Use an **eyepiece graticule** calibrated with a stage micrometer to measure sizes, and add a magnification or scale.

Prepared permanent slides are an acceptable alternative.

Making and examining a stem section

1. Cut several very thin transverse sections with a sharp scalpel or razor blade, cutting away from your fingers, and keep them in water.

   Thin sections are only one or two cells thick, so light passes through and the cells do not overlap. Water stops them drying out.

2. Stain the sections, for example with toluidine blue O, which turns lignified walls a green-blue and other walls a pink-purple, or with phloroglucinol and acid, which stains lignin red.

   Staining lignin picks out the xylem and sclerenchyma, the two tissues with thick lignified walls.

3. Rinse, mount the thinnest section in a drop of water under a coverslip, and find a vascular bundle at low power.

4. Identify the tissues by their position and appearance: large empty lignified vessels on the inner side (xylem), small thin-walled cells outside them (phloem), and a group of small, very thick-walled lignified cells at the outer edge (sclerenchyma).

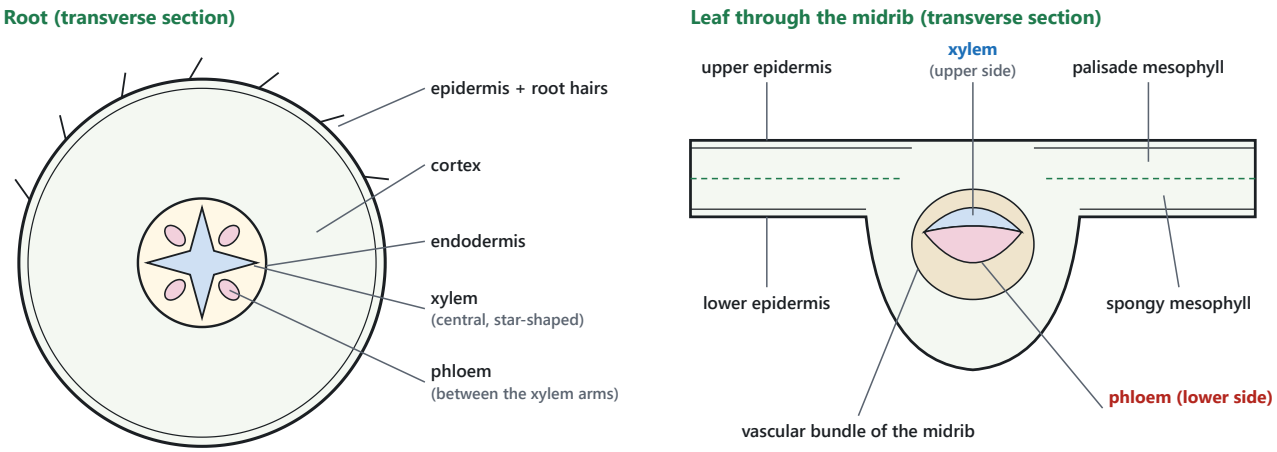
5. Draw a low-power plan showing the outlines of the tissues only, with no individual cells, using clear continuous lines and ruled label lines. State the magnification.

Stains and what they show

- **Toluidine blue O** stains **lignified** walls (xylem, sclerenchyma) **blue-green** and unlignified walls (phloem, parenchyma) **pink-purple**. Staining adds contrast, because thin unstained sections absorb very little light.
- (Other stains you may meet: **phloroglucinol** with acid stains lignin pink-red; **iodine** stains starch grains blue-black.)

Identifying the tissues

- **Xylem vessels**: the **largest**, rounded, empty cells with thick (lignified) walls, on the inner side of each bundle in a stem and in the centre of a root.
- **Sclerenchyma fibres**: small cells with **very thick** walls and tiny lumens, in a cap on the outside of each stem bundle.
- **Phloem**: a patch of **small, thin-walled** cells between the xylem and the sclerenchyma; in high power you may see **sieve plates** and small, densely stained **companion cells**.



Root: xylem forms a central star with phloem between its arms. Leaf: in the midrib and veins, xylem is on the upper side and phloem on the lower side.

Drawing rules (both kinds of drawing)

- Sharp **pencil**, **clear continuous lines**, **no shading** or colouring.
- Draw **large** (use at least half the space) and keep the **proportions** right.
- Plan diagrams: tissue boundaries only, **no cells**.
- Label lines drawn with a **ruler**, ending **exactly on** the structure, not crossing each other, no arrowheads. Labels outside the drawing.
- Add a **title** and the **magnification** (or a scale bar).
- Draw what you **see**, not what the textbook shows.

Safety

- Razor blades and scalpels: cut **away** from your fingers onto a tile; hand back or put in a sharps container.
- Toluidine blue stains skin and clothing: gloves, wipe up spills.
- Never point a microscope mirror at direct sunlight.

WORKED EXAMPLE

Worked example: calibrating a graticule

Eyepiece graticule: 40 divisions line up with 0.1 mm (100 µm) on the stage micrometer.

One division = 100 ÷ 40 = **2.5 µm** at this objective.

A xylem vessel spans 18 divisions: diameter = 18 × 2.5 = **45 µm**.

Recalibrate if you change the objective lens.

MARK-SCHEME WORDING

- Label lines must not touch the cambium line or the wrong tissue (WBI12 Jan 2020 report).
- A labelled diagram of the stem is accepted as part of an answer comparing root and stem arrangements.

Often asked as: the full plan diagram has not been set in WBI12 papers from 2019 to 2026, but its skills have: labelling xylem, phloem and sclerenchyma on a stem micrograph, comparing tissue arrangement in a root and a stem, and a stained celery section.

COMMON MISTAKE

- Drawing individual cells in a **plan** diagram, or shading.
- Few candidates knew where the tissues are in a root or a stem (WBI12 Oct 2021 report): learn the stem order (sclerenchyma, phloem, cambium, xylem) and the central root core.
- Air bubbles mistaken for structures: lower the coverslip at an angle.

SPEC 4.7 Plant fibres, starch and sustainability

WHAT THIS SPEC POINT ASKS

Explain how using plant fibres and starch can help sustainability, including plant-based products that replace oil-based plastics.

What "sustainable" means

A resource is used **sustainably** if we can meet today's needs **without** using it up, so it is still **available to future generations**. Plants are **renewable**: more can be grown. Crude oil is **non-renewable** (finite): once used, it is gone.

Plant products that can replace oil-based ones

- **Plant fibres** (hemp, jute, flax, sisal, cotton) for rope, cloth, sacks and building materials, instead of nylon and polyester made from oil.
- **Bioplastics** made from **starch** (e.g. from maize, potatoes, cassava) or **cellulose** (wood pulp), instead of oil-based plastics such as polythene and PVC. Examples: thermoplastic starch, PLA (polylactic acid, made by fermenting sugars from maize starch or sugarcane), cellulose films.
- **Bioethanol** made by fermenting sugars from starch or sugarcane, used as fuel instead of petrol.
- Waste plant material (sugarcane bagasse, wheat bran, seed husks, avocado seeds) can be turned into plates, cutlery or packaging.

Why plant-based products are more sustainable

Plant-based	Oil-based
Renewable: more crops can be grown each year	**Non-renewable**: oil will run out
Biodegradable: broken down by decomposers (bacteria and fungi)	persist for hundreds of years in landfill and the sea
Roughly **carbon neutral**: the CO_2 released when they rot or burn was absorbed by photosynthesis as the crop grew	extracting and burning oil adds CO_2 that was locked away, **increasing atmospheric CO_2**
Less energy needed to make many of them	energy-intensive refining

The drawbacks (needed for "discuss" or "evaluate")

- Crops for fibres, plastics and fuel compete with **food crops** for land, water and fertiliser.
- Growing them needs land (risk of **deforestation**, monocultures that lower biodiversity), fertilisers (**eutrophication**) and pesticides, and machinery that burns fuel, so they are not fully carbon neutral.
- Bioplastics may be weaker, less water-resistant or more expensive. Many only break down in industrial compost, not in soil or the sea.
- Plant ropes are often weaker than synthetic ones and can rot.

EXAM TIP

Every "more sustainable" point needs a reason: "renewable, **so** more can be grown and it won't run out for future generations"; "biodegradable, **so** decomposers break it down"; "carbon neutral, **so** no net increase in

atmospheric CO_2". The word "sustainable" on its own scores nothing.

MARK-SCHEME WORDING

- Sustainable: "renewable / more can be grown / will not run out / available to future generations"; "biodegradable / broken down by decomposers"; "carbon neutral".
- Oil-based not sustainable: "non-renewable (finite)", "takes a long time to decompose", "adds to CO_2 / not carbon neutral".
- **Ignored**: "recyclable", "less pollution" with no detail, "sustainable" with no explanation, "more easily broken down" with no detail. "Not recyclable" is not credited for oil-based plastics.
- Packaging with antimicrobial plant extracts (e.g. oregano): "antimicrobial property reduces bacteria on the food", so a longer shelf life and fewer food-poisoning cases.

Often asked as: explain why a named plant product (hemp, wheat bran, avocado seed, a calabash bowl, wood packaging) is more sustainable than an oil-based one; a 6-mark comparison of bioethanol crops; percentage calculations from product data.

COMMON MISTAKE

- Giving "more sustainable" with no reason (WBI12 Jun 2023 report), or leaving out the **future generations** idea (WBI12 Oct 2019 report).
- Writing "less pollution" or "recyclable". Neither earns a mark.
- In a 6-mark evaluation, listing pros and cons without linking them to sustainability (capped at Level 1).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Bioethanol crops** 6-mark: compare water, fertiliser and pesticide needs and CO_2 output of crops such as hemp, rapeseed and soybean; consider the farmer, the population and the ecosystem; reach a supported conclusion (WBI12 Oct 2019).
- **Starch bioplastic** strength: more starch → higher tensile strength (more hydrogen bonds between starch molecules); more **glycerol** (a plasticiser) → lower tensile strength (WBI12 Jan 2026).
- Percentage of a percentage: e.g. 14% of 48 mg, then 43% of that answer, to 2 s.f. (WBI12 Oct 2020).

SPEC 4.8 Water and inorganic ions in plants

WHAT THIS SPEC POINT ASKS

Explain why plants need water and the inorganic ions nitrate, calcium and magnesium.

Roots take in water by osmosis and mineral ions mostly by **active transport** (using ATP) from the soil. Both travel up the plant in the **xylem**.

Water

- **Reactant in photosynthesis** (split in the light-dependent stage) and in **hydrolysis** reactions (e.g. starch → glucose).
- **Solvent**: substances dissolve in it so they can react and be **transported** (mineral ions in the xylem, sucrose in the phloem).
- **Turgor**: water entering vacuoles by osmosis makes cells **turgid**, which **supports** non-woody plants and leaves. Without enough water, cells become flaccid and the plant **wilts**. Turgor also opens stomata (guard cells).
- **Cooling**: evaporation of water from leaves (transpiration) cools them.

Inorganic ions

Ion	Used for	Deficiency symptoms
Nitrate (NO_3^-)	source of **nitrogen** to make **amino acids** → **proteins** (enzymes, membrane proteins), **nucleotides** → **DNA, RNA**, **ATP**, NADP, and **chlorophyll**	stunted growth; leaves turn yellow (**chlorosis**), starting with the **older** leaves
Calcium (Ca^{2+})	forms **calcium pectate** in the **middle lamella**, which holds neighbouring cells together, and in the wall matrix; also needed for membrane function	growing tips and **young** leaves die back or are deformed; weak walls; soft fruit
Magnesium (Mg^{2+})	part of every **chlorophyll** molecule, so needed for light absorption and photosynthesis; also activates some enzymes (ATP works bound to Mg^{2+})	yellowing between the veins, starting with **older** leaves; less photosynthesis, so slower growth

Older leaves show nitrogen and magnesium shortage first because the plant moves these ions out of old leaves into new growth. Calcium can't be moved once it is in a wall, so shortage shows in the **youngest** tissue.

EXAM TIP

Follow the chain to the end: "magnesium → **chlorophyll** → light absorbed → **photosynthesis** → glucose made → growth". "Magnesium is needed for photosynthesis" stops too early, and "calcium pectate" with no job scores nothing. Say what it is **for**: "calcium pectate **holds cells together** in the middle lamella".

CORE PRACTICAL

Recommended practical: plant mineral deficiencies

1. Grow seedlings of the same species, age and size in **culture solutions**: one complete (all ions, the **control**) and others each lacking one ion (e.g. no nitrate, no magnesium, no calcium).
2. Keep everything else the same: volume of solution, light, temperature, aeration of the solution, time.
3. Cover containers so that algae don't grow, and top up with distilled water.
4. After a few weeks, record **leaf colour**, number of leaves, shoot length, and **dry mass** (dry at 80–100 °C to constant mass) or chlorophyll content.
5. Use several plants per solution and compare means (with error bars or standard deviation).

Growing seedlings without one ion

1. Set up identical containers of mineral solution: one complete, and others each lacking one ion (no nitrate, no calcium, no magnesium). A container of distilled water can be added for comparison.

   The complete solution is the control: any difference in the other plants must be caused by the missing ion.

2. Put seedlings of the same species, age and size into each solution, supported so the roots are in the liquid. Cover the containers with foil or black paper.

   Darkness around the roots stops algae growing in the solution and competing for the ions.

3. Keep all the plants at the same light intensity and temperature, and bubble air through the solutions or replace them regularly.

   Roots need oxygen for the aerobic respiration that powers active uptake of ions.

4. After a few weeks, record the height, number and colour of leaves, root length and dry mass of each plant, and use several plants per solution for a mean.

MARK-SCHEME WORDING

- Calcium: "calcium pectate to {hold cells together / form the middle lamella / increase cell wall stability}". "Makes cell walls" and "pectate" alone are ignored. Higher-level answers add that calcium pectate holds microfibrils together in the wall matrix.
- Magnesium: "to make chlorophyll so {photosynthesis can occur / light energy is absorbed / glucose is made}". "Plant growth" and "chloroplasts" are ignored. Mg^{2+} activating enzymes is accepted.
- Nitrate: "to make {amino acids / proteins / DNA / RNA / nucleic acids / chlorophyll}", naming the compound **and** its role (e.g. "proteins for enzymes").

- Water: "for {hydrolysis / photosynthesis}", "as a solvent / to transport named substances", "for turgidity". "Osmotic effects" alone is ignored.
- Data questions: state the trend with both variables, compare with the **control**, and say **which** error bars overlap (no significant difference) or don't (significant difference).

Often asked as: explain why a plant needs Ca^{2+}, Mg^{2+} or NO_3^-; why carnivorous plants catch insects (nitrogen for amino acids and nucleic acids); calcium and fruit firmness; 6-mark questions on water stress, eggshell calcium fertiliser or hydroponic magnesium.

COMMON MISTAKE

- Describing deficiency symptoms when the question asks for the ion's **role** (WBI12 Jun 2019 report).
- Stopping at "amino acids" or "proteins" for nitrate with no further role (WBI12 Oct 2019 report).
- Writing "food" instead of glucose, or "calcium" with no link to calcium pectate.
- Not comparing with the control group in data questions (WBI12 Jan 2025 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Fewer Ca^{2+} ions (e.g. a chemical such as EDTA or verapamil that reduces uptake) → less calcium pectate → weaker walls and a weaker middle lamella (WBI12 Jan 2025).
- Ions enter roots by **active transport** (WBI12 Jun 2025).
- Water stress 6-mark: less water → stomata close → less CO_2 in → less photosynthesis; smaller roots take up fewer ions; less cellulose → thinner, weaker walls; lower growth and yield (WBI12 Jun 2024).
- Nitrogen-fixing bacteria in root nodules (e.g. alfalfa) supply the plant with nitrogen compounds (WBI12 Jan 2026).
- Percentage difference: follow the formula the question gives (one paper used (A − B) ÷ (A + B) × 100) (WBI12 Jun 2025).

SPEC 4.9 Core Practical 8: tensile strength of plant fibres

WHAT THIS SPEC POINT ASKS

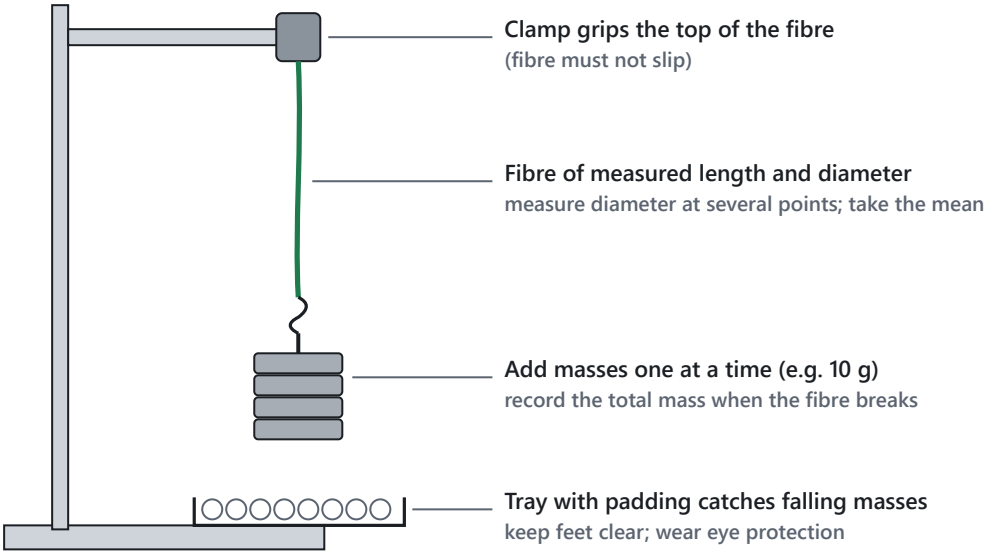
Measure the tensile strength of plant fibres.

Principle

Tensile strength is the **stretching force a material can withstand before it breaks, per unit cross-sectional area**:

tensile strength = breaking force (N) ÷ cross-sectional area (m²)

Units: N m^{-2} (= Pa), or N mm^{-2}. A hanging mass gives a force: **force (N) = mass (kg) × 9.81**. Dividing by area lets you compare thick and thin fibres fairly.



Set-up for measuring breaking load. Masses are added one at a time until the fibre snaps.

CORE PRACTICAL

Core practical 8: method in brief

1. Extract fibres (e.g. from celery, nettle, flax or banana stems) by scraping or by **retting** (soaking so microbes break down the soft tissue). Choose fibres with no visible damage.
2. Cut each fibre to the **same length** (e.g. 10 cm). Measure its **diameter** at several points with a micrometer or a microscope with a graticule, and take the mean.
3. Clamp the top of the fibre to a retort stand (or tie it round a bar), and hang a mass hanger from the bottom. Put a **tray with padding** underneath.
4. Add masses **one at a time** (e.g. 10 g), waiting a few seconds after each, until the fibre **breaks**. Record the total mass.
5. Repeat with at least three fibres of each type and calculate the mean breaking mass.
6. Convert to force (× 9.81, with mass in kg) and divide by the cross-sectional area (πr^2) to get tensile strength.

Measuring the breaking mass of a fibre

1. Extract fibres from the plant, for example by retting nettle or flax stems, or by pulling the vascular strands out of a celery stalk. Use fibres of the same length.

2. Clamp one end of the fibre firmly to a stand and tie a mass hanger to the other end. Place a tray of sand or a cushion underneath, and wear eye protection.

   When the fibre snaps the masses fall; the tray protects feet and bench, and goggles protect against a whipping fibre.

3. Add masses one at a time, gently and without dropping them, until the fibre breaks. Record the total mass, including the hanger, at the moment it breaks.

   Dropping a mass on would add a sudden extra force and give a falsely low value.

4. Repeat with several fibres of the same type and calculate a mean. Then repeat with other types of fibre.

   Natural fibres vary a lot, so repeats are needed to give a reliable mean and show the spread.

5. Convert to force: $F = m \times g$, with m in kg and $g = 9.81\ N\ kg^{-1}$. Measure the fibre's diameter with a micrometer or calibrated eyepiece graticule, find the area $A = \pi r^2$, and calculate tensile strength = $F \div A$.

Variables

- **Independent**: type of fibre (or plant species, age, treatment).
- **Dependent**: breaking mass / force, then tensile strength.
- **Control**: fibre length, diameter (or correct for it with area), the part of the plant it came from, its age and how fresh or dry it is, the size of each mass added, the time between additions.

Safety

Wear **eye protection** (a snapping fibre can whip up). Keep feet clear of falling masses and use the padded tray. Take care with knives or scalpels when extracting fibres.

WORKED EXAMPLE

Worked example: tensile strength from breaking mass

A fibre breaks when the total mass is 450 g. Its mean diameter is 0.80 mm.

Force = 0.450 kg × 9.81 = **4.41 N**.

Radius = 0.80 ÷ 2 = 0.40 mm = 4.0×10^{-4} m.

Area = $\pi r^2 = \pi \times (4.0 \times 10^{-4})^2 = \mathbf{5.03 \times 10^{-7}\ m^2}$.

Tensile strength = $4.41 \div 5.03 \times 10^{-7} = \mathbf{8.8 \times 10^{6}\ N\ m^{-2}}$ (2 s.f.).

Use the **radius**, not the diameter, and convert mm to m **before** squaring.

Worked example: tensile strength

1. A fibre of diameter 0.10 mm breaks when the total mass is 400 g.
2. Force: $F = 0.400\ kg \times 9.81\ N\ kg^{-1} = 3.92\ N$.
3. Area: $r = 0.05\ mm = 5.0 \times 10^{-5}\ m$, so $A = \pi \times (5.0 \times 10^{-5})^2 = 7.85 \times 10^{-9}\ m^2$.
4. Tensile strength $= 3.92 \div 7.85 \times 10^{-9} = 5.0 \times 10^{8}\ Pa = 500\ MPa$.

Dividing by area lets fibres of different thickness be compared fairly: a thick strand can hold more mass yet still be a weaker material.

The simulation at the bottom of this page lets you run the practical. The breaking loads are simulated but behave like real fibres, which vary from one sample to the next.

Errors and improvements

- Fibre slips in the clamp or breaks **at** the clamp → wrap the end or tie it round a smooth bar; ignore results where it breaks at the clamp.
- Mass steps too large → the true breaking load lies between two values; use **smaller masses** near the expected break.
- Masses added with a jerk add extra force → lower them gently.
- Fibres vary in diameter and have weak spots → measure diameter at several points, use more fibres, calculate means.
- Fibres dry out over time → test them all fresh, or all at the same time after extraction.

COMMON MISTAKE

- Reporting the **breaking mass** as "tensile strength". It is only a breaking load until you convert to force and divide by area. Longer celery strings may seem "stronger" simply because they were thicker.
- Assuming celery "strings" are xylem. They are mainly strands of thick-walled supporting cells (collenchyma) running alongside the vascular bundles, so a dye test that stains the xylem doesn't show that the whole string is xylem.

MARK-SCHEME WORDING

Not examined as a practical in WBI12 papers from 2019 to 2026. Tensile strength appears in explanations instead: why a fibre is strong (cellulose, hydrogen bonds, microfibrils at different angles, lignin), bioplastic strength data (starch and glycerol), and water stress reducing cellulose and therefore wall strength.

Often asked as: link tensile strength to microfibril structure; interpret strength data with error bars.

SPEC 4.10 Conditions bacteria need to grow

WHAT THIS SPEC POINT ASKS

Know what conditions bacteria need for growth.

Bacteria reproduce by **binary fission**; in good conditions some divide every 20–30 minutes. Each condition below matters because it affects the enzymes and the processes the cell needs to grow and divide.

Condition	Why it is needed
Nutrients	a **carbon source** (e.g. glucose) for **respiration** and to build molecules; a **nitrogen source** (amino acids, nitrate) to make **proteins** and nucleic acids; mineral ions (e.g. phosphate for DNA and ATP). Agar or broth provides these.
Water	solvent for reactions and transport; needed for **hydrolysis** reactions; bacteria in a dry place stop growing.
Suitable temperature	an **optimum** temperature for their **enzymes**. Too cold → slow reactions (little kinetic energy); too hot → enzymes **denature**.
Suitable pH	each bacterium's enzymes have an optimum pH; far from it the active sites change shape (ionic and hydrogen bonds disrupted).
Oxygen (some species)	**obligate aerobes** need O_2 for **aerobic respiration**. **Obligate anaerobes** are killed or inhibited by O_2. **Facultative anaerobes** grow with or without it.

- Photosynthetic bacteria (e.g. cyanobacteria) need **light**, CO_2 and mineral ions instead of a supply of glucose.
- In school labs cultures are grown at **25 °C** or below, **not 37 °C**, so that bacteria which grow well at body temperature (likely human pathogens) are not encouraged.

MARK-SCHEME WORDING

- Each condition needs a reason: "oxygen for (aerobic) respiration" (or "no oxygen" for obligate anaerobes); "glucose for respiration / amino acids for protein synthesis"; "optimum temperature for

{enzyme / metabolic} reactions" (or "so enzymes are not denatured"); "optimum pH for … reactions"; "water for {hydrolysis / as a solvent}".

- Photosynthetic bacterium: mineral ions (nitrate, phosphate), pH, light. Glucose, temperature, "nutrients" and "moisture" are ignored there.
- Oxygen graph: "as oxygen increases the growth rate of A decreases whereas B increases"; "B is an (obligate) aerobe"; "B levels off because another factor is limiting"; "high oxygen inhibits A (an anaerobe)".

Often asked as: explain the conditions needed (4 marks); describe and explain growth-rate graphs against oxygen, salt (NaCl) concentration or temperature; bacterial survival on wooden and plastic chopping boards.

COMMON MISTAKE

- Listing conditions with no reasons: the commonest reason marks are lost (WBI12 Oct 2020, Jan 2021, Jun 2023 reports).
- Saying high temperature "denatures the bacteria". It denatures their **enzymes** (WBI12 Jan 2021 report).
- Giving **plant** growth conditions, or glucose for a photosynthetic bacterium.
- Describing each bar separately instead of giving the overall trend first (WBI12 Oct 2022 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Thermophiles** have heat-stable enzymes with a high optimum temperature; **psychrophiles** grow best in the cold (WBI12 Jan 2023).
- High **salt** lowers the water potential outside the cell, so water leaves the bacterium by osmosis and growth slows (WBI12 Jun 2022).

SPEC 4.11 Plants as sources of antimicrobial and other medicinal substances

WHAT THIS SPEC POINT ASKS

Know that substances made by plants can have antimicrobial and other therapeutic (medicinal) properties.

Why plants make these chemicals

Plants can't move away from pathogens or herbivores. Many make chemicals that **kill or slow the growth of microorganisms** (antimicrobials: antibacterial and antifungal) or deter animals (e.g. bitter **tannins**). Humans can extract, purify and use these, or copy their structure.

Examples

Plant	Substance	Use
Willow bark	salicylic acid → modified to **aspirin** (acetylsalicylic acid)	pain relief, reducing fever and inflammation, preventing blood clots
Cinchona bark	**quinine**	treating malaria
Foxglove (*Digitalis*)	**digitalis** (digoxin)	heart failure and dropsy (4.13)
Pacific **yew**	**paclitaxel** (taxol)	some cancers
Sweet wormwood	**artemisinin**	malaria
Garlic, mint, tea tree, thyme, oregano	antimicrobial compounds (e.g. allicin in garlic)	tested on bacteria in CP9; antimicrobial packaging

From plant to medicine

1. Find a plant used in traditional medicine or one that resists disease.
2. **Extract** the active substance and **purify** it, so the **dose** can be measured and controlled.
3. Work out its structure, so it can be **synthesised** in a lab: no need to harvest huge amounts of a wild plant, and chemists can modify it to work better or have fewer side effects.
4. **Test** it for safety and effectiveness (4.13).

This is one reason to **conserve biodiversity**: plants not yet studied, especially endemic ones in threatened habitats, may contain useful drugs.

MARK-SCHEME WORDING

- Zone of inhibition data: "extract {was effective / had antimicrobial properties} against all three bacteria"; "prevented growth of / killed bacteria around the disc"; "higher concentration → larger zone".
- Comparing two extracts: say which is better **against which bacterium**, then whether the SDs or error bars overlap (no significant difference) or not, then a method point (control disc, same volume and concentration of extract).
- Largest possible difference between two zones: upper limit of the bigger (mean + range or SD) minus lower limit of the smaller.
- A gel with no extract used in a wound trial acts as a **control** (like a placebo) for comparison. This is a **control**, not a controlled variable.

Often asked as: interpret zone-of-inhibition data for plant extracts; 6-mark data on ulcer treatments with antimicrobial plant extracts; wound-healing trials; calculate a standard deviation from zone diameters.

COMMON MISTAKE

- Calling penicillin a plant product. It comes from a **fungus**.
- Confusing a **control** (a group or disc without the active substance) with a **controlled variable** (WBI12 Jun 2023 report).
- Saying a standard deviation "proves reliability", or that overlapping SDs show "accuracy". Overlap means **no significant difference** (WBI12 Jun 2023, Jan 2024 reports).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Standard deviation** by hand, $s = \sqrt{(\Sigma(x - \bar{x})^2 \div (n - 1))}$ (formula given). SD shows how spread out the values are about the mean, not a correlation (WBI12 Oct 2024).
- Named plant chemicals in stems: **tannins** (bitter, deter herbivores), **flavonoids** (antibacterial), fucoxanthin from seaweed (WBI12 Jun 2021, Jan 2022).

SPEC 4.12 Core Practical 9: antimicrobial properties of plants and aseptic technique

WHAT THIS SPEC POINT ASKS

Investigate whether plant extracts have antimicrobial properties, using aseptic techniques to handle bacteria safely.

Principle

Paper discs soaked in a plant extract are placed on agar covered evenly with bacteria. If the extract kills bacteria or stops them growing, it diffuses into the agar and leaves a **clear zone** (zone of inhibition) with no growth around the disc. A bigger zone suggests a more effective antimicrobial.

CORE PRACTICAL

Core practical 9: method in brief

1. Wash hands and **disinfect the bench** (e.g. 1% Virkon left for 10 minutes). Close windows and doors.
2. **Grind** each plant material (e.g. garlic, mint) with a pestle and mortar in the same volume of **ethanol** (to dissolve the active substances). Leave, then filter or decant the extract.
3. Soak sterile paper discs (all the same size) in each extract for the **same time**; let the ethanol **evaporate** on a sterile surface.
4. Mark the **base** of an agar plate already seeded with a lawn of bacteria (e.g. *Bacillus subtilis*, *Micrococcus luteus*) into quarters.

5. Lift the lid only slightly and use **sterile forceps** (flamed and cooled) to place one disc per quarter. Include a **control disc soaked in ethanol only** (evaporated).
6. **Tape the lid at two points** (not all the way round), **invert** the plate and incubate at **25 °C** for **24–48 hours**.
7. Measure the **diameter** of each clear zone across two directions at right angles, take the mean, and calculate the area. Repeat with more plates and find mean values.

Testing plant extracts on a bacterial lawn

1. Crush a known mass of plant material (for example garlic, mint or ginger) with ethanol using a pestle and mortar, then filter. Each extract should be made the same way.

   Many antimicrobial substances in plants dissolve better in ethanol than in water.

2. Using aseptic technique, spread a known volume of a bacterial culture evenly over an agar plate to make a lawn.

3. Soak sterile paper discs of the same size in each extract, and one disc in ethanol only as a control. Let the discs dry, then place them on the agar with sterile forceps, well spaced apart.

   The ethanol must evaporate, otherwise it would kill bacteria itself. The control disc shows whether any killing is due to the solvent rather than the plant.

4. Tape the lid on with two short strips (not all the way round), label the base, and incubate upside down at 25 °C for 24 to 48 hours.

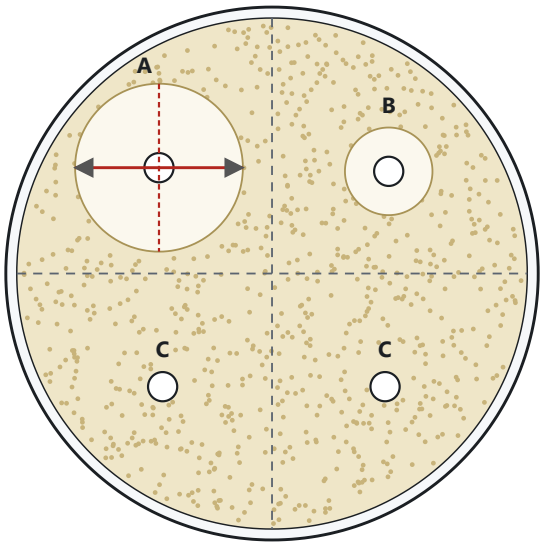
   Partial taping stops the lid coming off but lets oxygen in, which discourages anaerobic bacteria, many of which are pathogens. Upside down stops condensation dripping onto the agar.

5. Measure the diameter of the clear zone around each disc, across two diameters at right angles, and calculate the area of the zone with πr^2. A larger clear zone means a more effective antimicrobial at that concentration.

Aseptic technique and why each step matters

Step	Reason
Disinfect bench; wash hands	kill microbes that could contaminate the culture
Work near a lit **Bunsen burner**	rising hot air carries airborne microbes **away** from the plate

Step	Reason
Flame loops and forceps; **flame the neck** of broth bottles when opening and closing	kill microbes on surfaces that touch the culture
Sterile (autoclaved or single-use) plates, agar, pipettes, spreaders	no other microbes present at the start
Lift the lid only slightly and briefly	fewer airborne microbes fall in
Tape the lid at two points, not sealed all round	lets **oxygen** in, so harmful **anaerobic** bacteria don't grow; the plate can't be opened accidentally
Incubate **upside down**	condensation doesn't drip onto the agar and spread colonies
Incubate at **25 °C**, not 37 °C	avoids growing **human pathogens**
Never open plates after incubation; **autoclave** (121 °C) before disposal	protect people from any pathogens that grew



A, B = discs soaked in two plant extracts
C = control discs (solvent only, e.g. ethanol, evaporated)

Clear zone = no bacterial growth (inhibition)
Measure the diameter across two directions at right angles and take the mean (red lines on zone A)

Area of zone = πr^2, where r = mean diameter ÷ 2

Larger zone → more effective antimicrobial
(if concentration and diffusion are the same)

Lawn of bacteria spread evenly over the agar
Quarters marked on the BASE of the dish

Lid taped at two points (not sealed); dish inverted; incubated at 25 °C for 24–48 h

Discs A and B carry two plant extracts; C are solvent-only controls. Measure each clear zone across two diameters and take the mean.

KEY IDEA

Aseptic technique protects you from the bacteria and protects the culture from contamination: disinfect the bench, flame the inoculating loop until red hot and let it cool, flame the neck of culture bottles, lift the lid of a

plate only a little, work close to a lit Bunsen burner, and sterilise all used cultures and equipment in an autoclave (about 121 °C for 15 minutes) before disposal.

Variables and controls

- **Independent**: plant extract (or its concentration). **Dependent**: diameter (or area) of the clear zone.
- **Control**: same bacterium and volume of culture, same agar depth, same disc size, same volume and concentration of extract, same soaking time, same incubation temperature and time.
- The **ethanol-only disc** shows the solvent and the disc themselves don't stop growth. An **antibiotic disc** can be added as a positive control for comparison.

WORKED EXAMPLE

Worked example: area of a clear zone

Zone diameters measured across two directions: 18 mm and 20 mm. Mean = **19 mm**.

r = 19 ÷ 2 = 9.5 mm.

Area = πr^2 = π × 9.5^2 = **284 mm^2** (3 s.f.).

If the question asks for the area of clear agar only, subtract the disc: a 6 mm disc has area π × 3^2 = 28.3 mm^2, leaving 283.5 − 28.3 = 255 mm^2.

WORKED EXAMPLE

Worked example: standard deviation of zone diameters

Three plates give zone diameters of 7, 9 and 11 mm. Mean $\bar{x}$ = 27 ÷ 3 = **9 mm**.

$(x - \bar{x})$: −2, 0, +2. Squared: 4, 0, 4. $\Sigma(x - \bar{x})^2 = 8$.

$s = \sqrt{(8 \div (3 - 1))} = \sqrt{4}$ = **2.0 mm**. Remember the **n − 1** and the **square root**.

Limitations

- Different compounds have different-sized molecules, so they **diffuse** through agar at different rates: a small zone doesn't always mean a weak antimicrobial.
- The concentration of active substance in each extract is unknown and may differ.
- Zones may be irregular (measure two diameters, or use area); discs placed too close overlap.
- An agar plate is not skin or a gut: an extract that works here may be broken down in the body or be toxic.

MARK-SCHEME WORDING

- The method appears in question stems: agar plate spread with bacteria, discs soaked in extract, incubated at 25 °C for 24 h, zones measured.
- Planning answers credit: sterile equipment / aseptic technique; a range of concentrations; discs soaked in extract; a bacterial lawn; incubation (at least **24 hours**); the dependent variable as "the **diameter** of the clear zone" (not just "the clear zone").

Often asked as: aseptic technique itself was not asked in WBI12 papers from 2019 to 2026, but zone-of-inhibition data from this practical appear often (see 4.11).

COMMON MISTAKE

- "Aseptic technique means wearing gloves." It means preventing **contamination** of the culture and of people.
- Sealing the lid all the way round (encourages anaerobes), or incubating at 37 °C.
- Forgetting the solvent-only control disc.

SPEC 4.13 How drug testing developed: Withering to modern trials

WHAT THIS SPEC POINT ASKS

Know how drug testing changed from historical methods to today's protocols, including William Withering's digitalis soup, double-blind trials, placebos and the three phases of clinical trials.

William Withering and digitalis

- In the 1770s–80s, Withering, an English doctor, heard of a traditional herbal remedy (a "soup" of many plant ingredients) that helped patients with **dropsy** (swelling caused by fluid build-up, often from heart failure).
- He worked out that **foxglove** (*Digitalis purpurea*) was the **active ingredient**.
- Over about ten years he gave foxglove preparations (e.g. dried, powdered leaves) to more than 150 patients, **changing the dose** and **recording the results**. Too little did nothing; too much was toxic (sickness, slowed or irregular heartbeat); the right dose strengthened the heartbeat and removed the fluid.
- Today the drug from foxglove is **digoxin**, still used for some heart conditions.

What was missing from Withering's method

- No **pre-clinical** tests (cells, tissues, animals) before treating people.
- Tested straight on **ill patients**, not healthy volunteers first.
- No **control group** or **placebo**; not **randomised** or **blind**.

- The amount of active substance in each leaf varied, so doses were not standardised.
- Small numbers, no statistics, no independent checking, no formal consent.

Modern drug testing

1. **Pre-clinical testing**: computer modelling, then tests on human **cells and tissues** (in vitro), then on **animals**, to see whether the drug works, how toxic it is, and how it is absorbed, broken down and excreted.
2. **Clinical trials** in three phases (table below).
3. Data are reviewed independently between phases. If the results are good, the drug is **licensed**, and its use is still monitored afterwards.

Phase	Who	How many	What it tests
1	**healthy volunteers**	small group (tens)	**safety**, **side effects**, how the body handles the drug; a range of **doses** starting low
2	**patients with the disease**	a few hundred (about 100–500)	whether it **works**; the best **dose**; more on side effects; often against a placebo
3	**patients with the disease**	thousands (about 1000–3000)	whether it is **more effective than the existing treatment** (or a placebo); how common side effects are; usually **randomised, double-blind**

Stages in testing a new drug

1. **Pre-clinical testing:** the drug is tested on cells and tissues grown in the laboratory, then on animals such as rodents.

   This checks the drug acts as expected in living tissue, shows how it is absorbed and broken down in a whole organism, and screens out toxic compounds before any human takes them.

2. **Phase 1:** a small number of healthy volunteers are given the drug, starting with very low doses.

   The aim is safety: side effects, a safe dose range, and how the body absorbs, distributes and excretes the drug.

3. **Phase 2:** a few hundred patients who have the disease are treated.

   This tests whether the drug actually works against the disease, finds the best dose and records side effects in people who are ill.

4. **Phase 3:** a large trial with thousands of patients, usually randomised and double-blind, comparing the new drug with a placebo or with the best existing treatment.

   Large numbers allow statistical tests to show whether any improvement is significant, and reveal rarer side effects.

5. If all goes well the results are reviewed by a regulatory body, which may license the drug. Its safety continues to be monitored after it is in use.

Placebos and blinding

- A **placebo** looks identical to the drug (same colour, shape, taste) but has **no active ingredient**. It is given to the **control group** so that any improvement caused by the drug can be separated from the **placebo effect** (patients improving because they believe they are being treated).
- When an effective treatment already exists, the control group gets the **existing drug** instead, because withholding treatment would be unethical (e.g. for burns or cancer).
- **Single-blind**: only the patients don't know which treatment they get.
- **Double-blind**: **neither the patients nor the doctors/scientists** know who gets the drug and who gets the placebo (or existing drug) until the end. This removes **bias** from both sides: doctors can't treat or assess patients differently, and patients' expectations can't affect the results.
- Patients are **randomly** assigned to groups, and groups are matched for age, sex, health and severity of disease.

Withering compared with modern protocols

Withering's approach	Modern protocol
Tested directly on sick patients	Tested on cells, tissues and animals first, then on healthy volunteers
Crude plant extract with an uncontrolled amount of active ingredient	Pure active ingredient, so the dose is exact
Dose found by trial and error on patients, with some poisoned	Safe dose range established in phase 1 before patients are treated
No control group or placebo; patients and doctor knew the treatment	Randomised, double-blind, placebo-controlled trials
Small numbers of patients, no statistical analysis	Hundreds to thousands of patients and statistical tests of significance
Similar: careful observation, recording of side effects, gradual adjustment of the dose, and publication of the results	Similar: the same principles, now required by law and checked by regulators

EXAM TIP

Answer in context. If the question names a disease or a drug (plague, burns, headaches), use it in every phase: "Phase 2: a few hundred **patients with burns**…". Generic trial essays are capped at Level 1–2.

MARK-SCHEME WORDING

- Phases: pre-clinical on {human cells / animals}; Phase 1 on **healthy volunteers** (side effects, safety); Phase 2 on a **small group (100–500) of patients** (dose, and whether it works); Phase 3 on a **large group (1000–3000) of patients**, compared with a placebo or the current drug; double-blind; a statistical test; review between phases.
- Phase roles: "to test for side effects", "to find the {safe / minimum / effective} dose", "to see if it is more effective than {existing treatments / a placebo}". "Check for safety" or "see if it works" on their own are ignored. Side effects are **not** credited for cell or animal tests.
- Double-blind: "neither the {patient} nor the {doctor / scientist} knows whether they are receiving the {drug / placebo / existing treatment}".
- Finding a dose: "give a range of doses / increase the dose", "the lowest effective dose / highest dose without serious side effects".
- Withering: the active ingredient was **digitalis** (digoxin accepted). Modern trials know the active ingredient and its dose, test first on healthy volunteers, start with the lowest dose, and use three phases.
- For a burns treatment, a placebo group is ignored: compare with the **current hospital treatment**.

Often asked as: describe a full trial for a named drug (6-mark); roles of phases 1 and 2; compare phases 2 and 3; explain double-blind; compare Withering's method with modern testing; evaluate trial data (sample size, missing data, no information on age or sex, subjective measures such as pain).

COMMON MISTAKE

- Describing the three phases when the question asks **how** a double-blind trial works, or describing a **single**-blind trial (WBI12 Oct 2020, Oct 2021 reports).
- Wrong group sizes, or putting patients in Phase 1.
- Writing "safe" or "patients" unqualified; calling an existing treatment a "placebo".
- Saying Withering tested garlic or another plant. He used **foxglove**.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Patient numbers per phase (about 100–500 in Phase 2; 1000–3000 in Phase 3) are credited (WBI12 Jun 2019).
- In a trial, the **current drug** is the control in a later stage, not the placebo (WBI12 Oct 2023).
- A plague-treatment 6-mark reached Level 3 only with the disease context, three phases, double-blind, a placebo **and** either a significance test or an in-vitro test on the bacterium (WBI12 Jun 2019).

- Spotting missing data (no results for one drug at some doses) as an evaluation point (WBI12 Jan 2023).

Quick checks

QUICK CHECK

Which structure links the cytoplasm of two neighbouring plant cells?

- Middle lamella
- Plasmodesma
- Tonoplast
- Pit

Plasmodesmata are channels filled with cytoplasm passing through both cell walls. A pit is only a thin area of wall that water can cross, and the middle lamella is the calcium pectate layer that glues the walls together, so neither connects the cytoplasm.

QUICK CHECK

Why can cellulose form microfibrils but starch cannot?

- Cellulose has 1,6 glycosidic bonds that cross-link the chains
- Starch is made of β-glucose, which cannot form hydrogen bonds
- Cellulose chains are straight, so parallel chains can be held together by many hydrogen bonds
- Cellulose chains coil into a helix that locks into neighbouring helices

Alternate β-glucose units are inverted, giving straight chains whose –OH groups form hydrogen bonds with neighbouring chains. Cellulose has only 1,4 bonds, and it is starch, not cellulose, that is made of α-glucose and coils into a helix.

QUICK CHECK

Which feature is shared by xylem vessels and sclerenchyma fibres but not by phloem sieve tubes?

- End walls perforated by sieve plates
- Transport of water and mineral ions
- Living contents supported by companion cells
- Lignified secondary cell walls

Both xylem vessels and sclerenchyma fibres have thick lignified walls and are dead. Phloem walls are cellulose only. Transporting water is a job for xylem alone, so that option is the commonest trap.

QUICK CHECK

A plant shows yellowing between the veins of its older leaves. Which ion is most likely to be lacking?

- Magnesium
- Calcium
- Nitrate
- Sodium

Magnesium is part of chlorophyll, and a shortage causes yellowing between the veins of older leaves first. Calcium shortage shows at the growing tips and young leaves. Nitrate shortage causes general yellowing of older leaves with stunted growth rather than yellowing confined to the areas between the veins.

QUICK CHECK

A fibre with a cross-sectional area of 2.0×10^{-8} m² breaks under a force of 8.0 N. What is its tensile strength?

- 1.6×10^{-7} Pa
- 4.0×10^{7} Pa
- 4.0×10^{8} Pa
- 2.5×10^{-9} Pa

Tensile strength = force ÷ area = $8.0 \div 2.0 \times 10^{-8} = 4.0 \times 10^{8}$ Pa (400 MPa). Multiplying instead of dividing gives 1.6×10^{-7} Pa, and slipping a power of ten gives 4.0×10^{7} Pa.

QUICK CHECK

Why is a phase 3 drug trial carried out double-blind?

- So that the patients do not suffer side effects
- So that neither patients nor doctors can bias the results by knowing who has the drug
- So that the drug can be compared with animal results
- So that a smaller number of patients is needed

If either the patients or the doctors know who has the real drug, their expectations can affect what is reported and recorded. Blinding does not prevent side effects, and phase 3 trials still need thousands of patients.

QUICK CHECK

In an antimicrobial investigation, why is one disc soaked in ethanol only?

- To provide nutrients for the bacteria
- To sterilise the agar around the other discs
- To make the clear zones larger
- To show that any clear zones are caused by the plant extract rather than the solvent

The ethanol-only disc is the control. If it also produced a clear zone, the solvent had not evaporated and the results could not be credited to the plant substances.